# SSD MEMORY DESIGN

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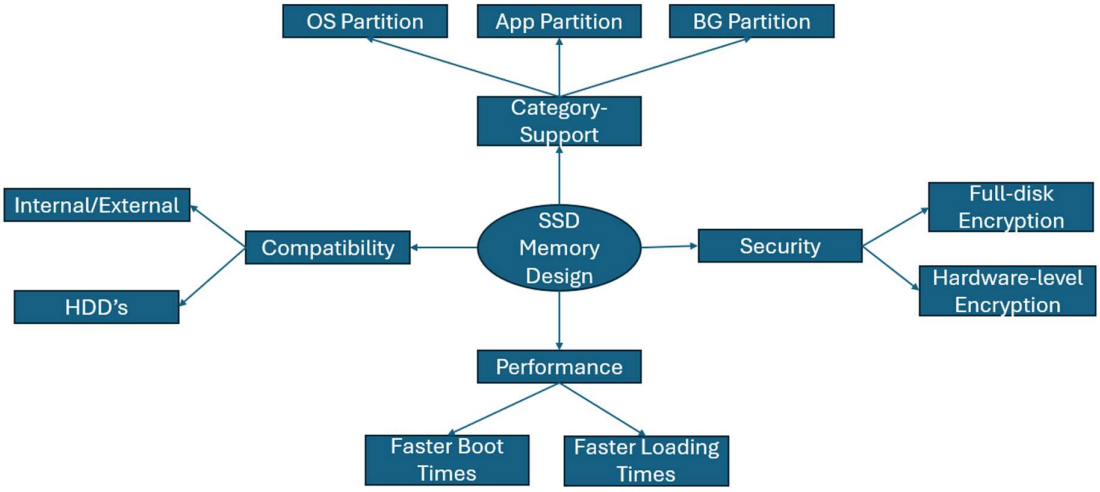
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# Introduction

**This SSD Memory Design is designed to improve performance on both internal and external SSDs. It is also designed to improve performance of older hardware such as internal and external HDDs. This SSD memory design was developed to allow for quicker loading times and boot times on modern computing systems that require performance and reliability. This SSD memory design does not require any direct hardware changes and can be implemented quickly through OS and minor firmware updates. This design will explain the category-based memory, the workflow and the trade-offs with migrations.**

**This SSD Memory Design cannot promise perfect performance or reliability. However, it can promise to improve performance through loading and boot times on modern computing systems. It can also provide compatibility with HDDs both externally and internally as this memory design is heavily managed by the operating system. It is important to remember that this memory design may be less effective on HDD's compared to SSDs. Make sure to adapt any features to the system's need.**




**The spider diagram above shows a visual layout of the SSD Memory Design. It shows 2-3 features of each principle involved in the design. This diagram does not show all features, and more detail can be found in the sections below.**

# SSD Memory Design Category-Based Memory

- **Category-based memory in an SSD or HDD involves splitting the data into partitions.**
- **This can be done by the operating system as they already manage partitions on hard drives and would just need to be configured to sort data based on category**
- **These partitions could be Operating System, Apps and Background.**
- **Files and processes can be tagged by the operating system to ensure that there is no confusion between what category they belong to.**
- **This also means that internal hard drives can be sorted by the OS for quicker access reducing load times and boot times.**
- **This design could be used to sort out external drives although this would require a system such as system partition for firmware and data partition for the data.**
- **Operating systems use two systems for allocating disk space**
- **The first system is fixed priority.**
- **Each partition on the internal drive would be dedicated a specific amount of space that cannot be changed by the user or adapted based on usage.**
- **This would also have free space which isn't used by any partition allowing the user to make their own custom partitions that they may use.**
- **For example, on an 1 TB SSD drive, OS partition would use 200 GB, background would use 70 GB, Application would use 500 GB. This means the user would have 230 GB to use for custom partitions.**
- **The second system is user-controlled priority system**
- **OS partition and Background Partitions would have the minimum amount of space needed with a small amount of extra space being given for edge cases.**
- **The application partition would be given the rest of the disk space for it to use for any apps or documents they may want to store on their device.**
- **Users would be able to create custom partition that would reduce the size used for the application partition meaning they have more control over their system.**
- **However, users can only make application partition sizes as small as the minimum requirement.**
- **This minimum requirement is calculated based on the applications and data stored within the partition.**
- **This SSD memory design for security would need to use full-disk encryption to ensure that all partitions are secured.**
- **Device encryption will not work for this design as we cannot just encrypt the OS partition as it would leave the application and background partitions vulnerable.**
- **DMA would use a 3-layer boot system first case, where it uses a boot header in the OS partition, second case, where it searches the whole OS partition, and third case, where it searches the other partitions on the system.**
- **The OS memory manager would need to be configured to manage the partitions.**
- **These changes could be rolled out in OS and firmware updates, but this varies.**

# SSD Memory Design Workflow

- **The workflow will show how the SSD design should work in a device and its connection to RAM, CPU, Cache and the GPU.**
- **For the first boot of the device, BIOS and other bootloader data are loaded from the hard drive without accessing any partitions as they don't exist yet**
- **The OS is then loaded from the SSD into RAM by the CPU and DMA by checking the whole disk space.**
- **After the OS is loaded, it sorts out all data on the device in the OS, App and Background partitions for quicker access.**
- **For the second boot, BIOS and other bootloader data are loaded from SSD into RAM to start up the device.**
- **This is done as the CPU and DMA can quickly access data from the OS partition leading to faster access times.**
- **OS is loaded from SSD into RAM to load the operating system into the device for the user to use.**
- **This is also done by the GPU and DMA as they can continue to quickly access data from the OS partition.**
- **This leads to faster boot times on devices especially those with NVMe SSDs for internal hard drives.**
- **The user starts a gaming application on the device after the second boot.**
- **For example, this gaming application could be GTA 6 or COD.**
- **The application is stored within the app partition where GTA-6 specific data is moved to RAM as instructions by the CPU and DMA**
- **The operating system then organises GTA-6 specific instructions into the application category to allow for the CPU to access it quicker.**
- **The most frequently used GTA-6 data and instructions are stored in L1 or L2 cache depending on the priority system that the device is configured with.**
- **The CPU can then access GTA-6 specific instructions quickly leading to faster loading times.**
- **GTA-6 3D Geometry, Images and Rendered Frames are moved to VRAM via the CPU's PCI bus.**
- **The operating system then organises the GTA-6 VRAM data into the 3 categories of Physics Model, Textures and Frame Buffers.**
- **L2 cache will prioritize Physics Models as it would be the heaviest workload of GTA 6.**
- **Each L1 cache and SMs/CUs will be dedicated to a specific category in VRAM with physics models being the main priority.**
- **Results are directly written to VRAM and then sent to the CPU for processing where they are then displayed on the user's monitor/screen.**

# SSD Memory Design Trade-Offs

- **This SSD memory design has its trade-offs just like any other system I have created.**
- **There may also be more implementation challenges although they should be relatively lighter compared to the GPU, RAM and Cache designs.**
- **I have identified 3 main trade-offs for the SSD Memory Design.**
- **The first trade-off is heat.**
- **This trade-off exists since DMA and CPU can access data from SSDs more quickly which can produce more heat.**
- **This means that without heat management the SSD can overheat causing issue for the user.**
- **This trade-off can be managed by hard drive-specific cooling methods such as passive cooling and general device cooling systems.**
- **The second trade-off is first-boot setup.**
- **This trade-off exists because the organised partitions have not been created by the operating system yet.**
- **This means that boot times may be slower as the CPU and DMA can take longer to access data from the SSD.**
- **This trade-off is acceptable as it is a one-time cost and external hard drives can completely avoid this trade-off as they don't rely on boots.**
- **The third trade-off is performance inequality.**
- **This trade-off exists since SSDs especially NVMes can perform more effectively due to faster read/write speeds compared to HDDs.**
- **This means that boot times and loading times may be slower on a device with an internal HDD compared to a device with an internal SSD.**
- **This trade-off is acceptable as HDDs are accepted by users as older hardware and it is managed as the memory architecture is also available for HDDs.**
- **The fourth trade-off is OS-Reliance**
- **This trade-off exists because the OS must perform partition management, key management, and category-sorting**
- **This means that the effectiveness of this SSD memory architecture may be limited by OS limitations.**
- **This trade-off is acceptable as operating systems act as the boundary between software and hardware, and it is also managed through other systems like DMA and the CPU.**